LABSHEET-1

Task 1: Basic Input and Output

a) Write a C++ program to display: Welcome to Data Structures Lab Course Code : BTCS209P.

```
#include <iostream>
using namespace std;

int main ()
{
   cout << "Welcome to Data Structures Lab"
<< endl;
   cout << "Course Code : BTCS209P";

   return 0;
}
```

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b) Write a C++ program to input two integers and display Sum, Difference, Product, Quotient and Remainder.

```
#include <iostream>
using namespace std;

int main()
{
   int a, b;
   cout << "Enter two integers: ";
   cin >> a >> b;

   cout << "Sum = " << a + b << endl;
   cout << "Difference = " << a - b << endl;
   cout << "Product = " << a * b << endl;
   cout << "Quotient = " << a / b << endl;
   cout << "Remainder = " << a % b << endl;

   return 0;
}
```

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Task 2: Decision Making Statements

a) Check whether a number is Even or Odd.

```
#include <iostream>
using namespace std;

int main() {
    int number;
    cout << "Enter a number: ";
    cin >> number;
    if (number % 2 == 0)
        cout << "Number is Even";
    else
        cout << "Number is Odd";
    return 0;
}
```

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b) Find the largest among three numbers.

```
#include <iostream>
using namespace std;

int main()
{
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;

    if (a >= b && a >= c)
        cout << "Largest = " << a;
    else if (b >= a && b >= c)
        cout << "Largest = " << b;
    else
        cout << "Largest = " << c;

    return 0;
}
```

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c) Check whether a year is a Leap Year.

```
#include <iostream>
using namespace std;
int main()
{
   int year;

   cout << "Enter a year: ";
   cin >> year;

   if ((year % 400 == 0) || (year % 4 == 0 &&
year % 100 != 0))
      cout << "Leap Year";
   else
      cout << "Not a Leap Year";

   return 0;
}
```

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Task 3: Looping Statements

a) Print numbers from 1 to 100 using for loop.

```
#include <iostream>
using namespace std;

int main()
{
    for (int i = 1; i <= 100; i++)
    {
        cout << i << " ";
    }

    return 0;
}
```

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b)Display multiplication table of a given number.

```
#include <iostream>
using namespace std;

int main()
{
   int number;
   cout << "Enter a number: ";
   cin >> number;

   for (int i = 1; i <= 10; i++)
   {
      cout << number << " x " << i << " = "
           << number * i << endl;
   }

   return 0;
}
```

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c) Find factorial using while loop.

```
#include <iostream>
using namespace std;
int main()
{
   int number;
   long long factorial = 1;
   cout << "Enter a number: ";
   cin >> number;

   int i = 1;
   while (i <= number)
   {
      factorial = factorial * i;
      i++;
   }

   cout << "Factorial = " << factorial;
   return 0;
}
```

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d) Generate Fibonacci series up to N terms.

```
#include <iostream>
using namespace std;
int main()
{
   int n;
   int first = 0, second = 1, next;
   cout << "Enter number of terms: ";
   cin >> n;
   cout << "Fibonacci Series: ";

   for (int i = 1; i <= n; i++)
   {
      cout << first << " ";
      next = first + second;
      first = second;
      second = next;
   }
   return 0;
}
```

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Task 4: Functions

a) Find square using a user-defined function.

```
#include <iostream>
using namespace std;

int square(int number)
{
   return number * number;
}
int main()
{
   int number;
   cout << "Enter a number: ";
   cin >> number;
   cout << "Square = " << square(number);

   return 0;
}
```

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b) Find maximum of two numbers using a function.

```
#include <iostream>
using namespace std;
int maximum(int a, int b)
{
   if (a > b)
      return a;
   else
      return b;
}
int main()
{
   int a, b;
   cout << "Enter two numbers: ";
   cin >> a >> b;
  cout << "Maximum = " << maximum(a, b);

   return 0;
}
```

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c) Check Prime number using a function.

```
#include <iostream>
using namespace std;

bool isPrime(int number)
{
   if (number <= 1)
      return false;

   for (int i = 2; i < number; i++)
   {
      if (number % i == 0)
         return false;
   }

   return true;
}
```

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```
int main()
{
   int number;

   cout << "Enter a number: ";
   cin >> number;

   if (isPrime(number))
      cout << "Prime Number";
   else
      cout << "Not a Prime Number";

   return 0;
}
```

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Task 5: Arrays

a) Read 10 integers into an array and display them.

```
#include <iostream>
using namespace std;
int main()  {
    int arr[10];
    cout << "Enter 10 integers: ";
    for (int i = 0; i < 10; i++)
    {
        cin >> arr[i];
    }
    cout << "Array elements are: ";

    for (int i = 0; i < 10; i++)
    {
        cout << arr[i] << " ";
    }
    return 0;
}
```

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b) Find largest and smallest elements.

```
#include <iostream>
using namespace std;

int main()
{
   int arr[10];

   cout << "Enter 10 integers: ";

   for (int i = 0; i < 10; i++)
   {
      cin >> arr[i];
   }

   int largest = arr[0];
   int smallest = arr[0];

   for (int i = 1; i < 10; i++)
```

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```
    {
        if (arr[i] > largest)
            largest = arr[i];

        if (arr[i] < smallest)
            smallest = arr[i];
    }

    cout << "Largest = " << largest << endl;
    cout << "Smallest = " << smallest;

    return 0;
}
```

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c) Calculate sum and average.

```
#include <iostream>
using namespace std;
int main() {
    int arr[10];
    int sum = 0;
    double average;
    cout << "Enter 10 integers: ";
    for (int i = 0; i < 10; i++)
    {
        cin >> arr[i];
        sum = sum + arr[i];
    }

    average = sum / 10.0;
    cout << "Sum = " << sum << endl;
    cout << "Average = " << average;

    return 0;
}
```

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d) Search an element using Linear Search.

```
#include <iostream>
using namespace std;

int main() {
   int A[10], n, data, i, flag = 0;

   cout << "Enter the no. of elements: ";
   cin >> n;
   if (n > 10) {
      cout << "Number of elements
cannot exceed 10." << endl;
      return 1;
   }
   cout << "Enter the array elements:\n";
   for (i = 0; i < n; i++) {
      cin >> A[i];
   }
```

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```
    cout << "Enter the Element/data to be
searched: ";
    cin >> data;

    for (i = 0; i < n; i++) {
        if (A[i] == data) {
            cout << "The element found at
index: " << i << endl;
            flag = 1;
            break;
        }
    }

    if (flag == 0) {
        cout << "Element not found" <<
endl;
    }

    return 0;
}
```

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LABSHEET-2

Task 1: Array Traversal:

Read N elements into an array and display the elements, indices, sum, and average.

```
#include <iostream>
using namespace std;
int main()  {
    int n;
    int arr[100];
    int sum = 0;
    cout << "Enter number of elements: ";
    cin >> n;
    if (n <= 0 || n > 100) {
        cout << "Invalid number of elements!";
        return 0;
    }
```

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```
cout << "Enter " << n << " elements:\n";

    for (int i = 0; i < n; i++) {
        cin >> arr[i];
        sum += arr[i];
    }
    cout << "\nArray Elements and Indices:\n";

    for (int i = 0; i < n; i++) {
        cout << "Index " << i << " = " << arr[i] << endl;
    }
    double average = (double)sum / n;
    cout << "\nSum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

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Task 2: Array Insertion:

Write a C++ program to insert an element at the beginning, end, and a given index. Display the array before and after insertion.

```
#include <iostream>
using namespace std;

int main() {
    int arr[100], n, choice, element, index;
    cout << "Enter number of elements: ";
    cin >> n;

    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
```

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```
  cout << "\nArray before insertion: ";
  for (int i = 0; i < n; i++) {
     cout << arr[i] << " ";
  }
  cout << "\n\n1. Insert at Beginning";
  cout << "\n2. Insert at End";
  cout << "\n3. Insert at Given Index";
  cout << "\nEnter your choice: ";
  cin >> choice;

  cout << "Enter element to insert: ";
  cin >> element;

  if (choice == 1) {

     for (int i = n; i > 0; i--) {
        arr[i] = arr[i - 1];
}
```

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```
    arr[0] = element;
    n++;
}
else if (choice == 2) {

    arr[n] = element;
    n++;
}
else if (choice == 3) {

    cout << "Enter index: ";
    cin >> index;

    if (index < 0 || index > n) {
        cout << "Invalid index!";
        return 0;
    }
```

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```
        for (int i = n; i > index; i--) {
            arr[i] = arr[i - 1];
        }
        arr[index] = element;
        n++;
    }
    else {
        cout << "Invalid choice!";
        return 0;
    }
    cout << "\nArray after insertion: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

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Task 3: Array Deletion:

Write a C++ program to delete an element from the beginning, end, and a given index. Display the array before and after deletion.

```
#include <iostream>
using namespace std;

int main() {
    int arr[100], n, choice, index;
    cout << "Enter number of elements: ";
    cin >> n;

    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
```

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```
// Display array before deletion
cout << "\nArray before deletion: ";
for (int i = 0; i < n; i++) {
    cout << arr[i] << " ";
}
cout << "\n\n1. Delete from Beginning";
cout << "\n2. Delete from End";
cout << "\n3. Delete from Given Index";
cout << "\nEnter your choice: ";
cin >> choice;

if (choice == 1) {

    for (int i = 0; i < n - 1; i++) {
        arr[i] = arr[i + 1];
    }
```

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```
    n--;
}

// Delete from End
else if (choice == 2) {

    n--;
}
else if (choice == 3) {

    cout << "Enter index to delete: ";
    cin >> index;

    // Check valid index
    if (index < 0 || index >= n) {
        cout << "Invalid index!";
        return 0;
    }
```

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```
        for (int i = index; i < n - 1; i++) {
            arr[i] = arr[i + 1];
        }

        n--;
    }
    else {
        cout << "Invalid choice!";
        return 0;
    }
    cout << "\nArray after deletion: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

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Task 4: Searching and Sorting:

Search an element using Linear Search and sort the array using Bubble Sort in ascending order.

```
#include <iostream>
using namespace std;
int main() {
    int arr[100], n, search, position = -1;
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter array elements: ";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }
    cout << "\nOriginal array: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
```

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```
    }
    cout << "\n\nEnter element to search: ";
    cin >> search;
    for (int i = 0; i < n; i++) {
        if (arr[i] == search) {
            position = i;
            break;
        }
    }
    if (position != -1) {
        cout << "Element found at index: " <<
position << endl;
    }
    else {
        cout << "Element not found!" << endl;
    }
    for (int i = 0; i < n - 1; i++) {
```

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```
        for (int j = 0; j < n - i - 1; j++) {

            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    cout << "\nArray after Bubble Sort (Ascending):
";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

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Task 5: Menu-Driven Array Operations:

Create a menu-driven program to display the array, insert, delete, search, sort, and exit. Test valid and invalid inputs.

```
#include <iostream>
using namespace std;
void display(int arr[], int n) {
    if (n == 0) {
        cout << "Array is empty!" << endl;
        return;
    }
    cout << "Array: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}
```

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```
void insertElement(int arr[], int &n) {
    int element, index;

    if (n >= 100) {
        cout << "Array is full!" << endl;
        return;
    }

    cout << "Enter element to insert: ";
    cin >> element;

    cout << "Enter index (0 to " << n << "): ";
    cin >> index;

    if (index < 0 || index > n) {
        cout << "Invalid index!" << endl;
```

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```
        return;
    }
    for (int i = n; i > index; i--) {
        arr[i] = arr[i - 1];
    }

    arr[index] = element;
    n++;

    cout << "Element inserted successfully!" <<
endl;
}
void deleteElement(int arr[], int &n) {
    int index;

    if (n == 0) {
        cout << "Array is empty!" << endl;
```

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```
        return;
    }
    cout << "Enter index to delete (0 to " << n - 1 <<
"): ";
    cin >> index;

    if (index < 0 || index >= n) {
        cout << "Invalid index!" << endl;
        return;
    }

    // Shift elements to the left
    for (int i = index; i < n - 1; i++) {
        arr[i] = arr[i + 1];
    }

    n--;
```

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```
    cout << "Element deleted successfully!" << endl;
}

// Function for linear search
void searchElement(int arr[], int n) {
    int element;
    bool found = false;

    cout << "Enter element to search: ";
    cin >> element;

    for (int i = 0; i < n; i++) {
        if (arr[i] == element) {
            cout << "Element found at index " << i << endl;
            found = true;
```

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```
            break;
        }
    }
    if (!found) {
        cout << "Element not found!" << endl;
    }
}
void sortArray(int arr[], int n) {

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {

            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
```

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```
            }
        }
    }
    cout << "Array sorted successfully!" << endl;
}
int main() {

    int arr[100];
    int n, choice;
    cout << "Enter number of elements: ";
    cin >> n;
    if (n < 0 || n > 100) {
        cout << "Invalid number of elements!" <<
endl;
        return 0;
    }
```

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```
cout << "Enter array elements: ";
for (int i = 0; i < n; i++) {
    cin >> arr[i];
}
do {
    cout << "\nARRAY MENU”<< endl;
    cout << "1. Display Array" << endl;
    cout << "2. Insert Element" << endl;
    cout << "3. Delete Element" << endl;
    cout << "4. Search Element" << endl;
    cout << "5. Sort Array" << endl;
    cout << "6. Exit" << endl;
    cout << " " << endl;
    cout << "Enter your choice: ";
    cin >> choice;
```

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```
switch (choice) {
    case 1:
        display(arr, n);
        break;

    case 2:
        insertElement(arr, n);
        break;

    case 3:
        deleteElement(arr, n);
        break;

    case 4:
        searchElement(arr, n);
        break;
```

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```
            case 5:
                sortArray(arr, n);
                break;

            case 6:
                cout << "Program exited successfully!" <<
endl;
                break;
            default:
                cout << "Invalid choice! Please enter 1 to
6." << endl;
        }
    } while (choice != 6);

    return 0;
}
```

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LABSHEET-3

Task 1: Create and Traverse a Singly Linked List:

Write a C++ program to create a singly linked list using user-entered data, display all nodes by traversing the list and count the total number of nodes.

```
#include <iostream>
using namespace std;
struct Node {
    int data;
    Node* next;
};
int main() {
    Node* head = new Node();
    Node* second = new Node();
    Node* third = new Node();
```

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```
    head->data = 10;
    second->data = 20;
    third->data = 30;

    head->next = second;
    second->next = third;
    third->next = nullptr;

    Node* temp = head;
    while (temp != nullptr) {
        cout << temp->data << " ";
        temp = temp->next
    }
    return 0;
}
```

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Task 2: Insert a Node :

Write a C++ program to insert a new node at the beginning, at the end, and at a given position in a singly linked list.

```
#include <iostream>
using namespace std;

int main() {
    struct Node {
        int data;
        Node* next;
    };
    Node* head = new Node();
    Node* second = new Node();
    Node* third = new Node();

    head->data = 10;
```

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```
second->data = 20;
third->data = 30;

head->next = second;
second->next = third;
third->next = nullptr;

cout << "Original List: ";
Node* temp = head;
while (temp != nullptr) {
    cout << temp->data << " ";
    temp = temp->next;
}
cout << endl;
Node* newNodeBeg = new Node();
newNodeBeg->data = 40;
```

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```
newNodeBeg->next = head;
head = newNodeBeg;

cout << "After Insertion at Beginning (40): ";
temp = head;
while (temp != nullptr) {
    cout << temp->data << " ";
    temp = temp->next;
}
cout << endl;
Node* newNodeEnd = new Node();
newNodeEnd->data = 50;
newNodeEnd->next = nullptr;

temp = head;
while (temp->next != nullptr) {
```

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```
    temp = temp->next;
}
temp->next = newNodeEnd;

cout << "After Insertion at End (50): ";
temp = head;
while (temp != nullptr) {
    cout << temp->data << " ";
    temp = temp->next;
}
cout << endl;
int P = 3;
Node* newNodePos = new Node();
newNodePos->data = 99;

temp = head;
```

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```
    for (int i = 1; i < P - 1; i++) {
        temp = temp->next;
    }
    newNodePos->next = temp->next;
    temp->next = newNodePos;
    cout << "After Insertion at Position 3 (99): ";
    temp = head;

    while (temp != nullptr) {
        cout << temp->data << " ";
        temp = temp->next;
    }
    cout << endl;

    return 0;
}
```

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Task 3: Delete a Node:

Write a C++ program to delete a node from the beginning, from the end, and from a given position in a singly linked list.

```
#include <iostream>
using namespace std;
struct Node {
    int data;
    Node* next;
};
int main() {
    Node* head = new Node();
    Node* second = new Node();
    Node* third = new Node();
    Node* forth = new Node();

    head->data = 10;
```

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```
second->data = 20;
third->data = 30;
forth->data = 40;

head->next = second;
second->next = third;
third->next = forth;
forth->next = nullptr;
if (head == nullptr) {
    cout << "Empty List" << endl;
} else {
    Node* temp = head;
    for (int i = 1; i < 3 - 1; i++) {
        temp = temp->next;
    }
    Node* temp1 = temp->next;
```

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```
    temp->next = temp1->next;
    cout << temp1->data << endl;
    delete temp1;
}
if (head == nullptr) {
    cout << "Empty List" << endl;
} else {
    Node* temp = head;
    while (temp->next->next != nullptr) {
        temp = temp->next;
    }
    Node* temp1 = temp->next;
    cout << temp1->data << endl;
    delete temp1;
    temp->next = nullptr;
}
```

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```
    if (head == nullptr) {
        cout << "Empty List" << endl;
    } else {
        Node* temp = head;
        head = head->next;
        cout << temp->data << endl;
        delete temp;
    }
    Node* temp = head;
    while (temp != nullptr) {
        cout << temp->data << endl;
        temp = temp->next;
    }

    return 0;
}
```

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Task 4: Search and Count:

Write a C++ program to search for a given element in a singly linked list.

```
#include <iostream>
using namespace std;
struct Node {
    int data;
    Node* next;
};
int main() {
    Node* head = new Node();
    Node* second = new Node();
    Node* third = new Node();
    Node* forth = new Node();
    Node* fifth = new Node();
    head->data = 10;
```

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```
second->data = 30;
third->data = 30;
forth->data = 40;
fifth->data = 50;

head->next = second;
second->next = third;
third->next = forth;
forth->next = fifth;
fifth->next = nullptr;

int pos = 1, count = 0, item;
item = 30;
Node* temp = head;

while (temp != nullptr) {
```

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```
        if (temp->data == item) {
            count++;
            cout << "Element found at position " << pos
<< endl;
        }
        temp = temp->next;
        pos++;
    }
    if (count > 0) {
        cout << "Total occurrences of " << item << " is:
" << count << endl;
    } else {
        cout << "Element not found" << endl;
    }
    return 0;
}
```

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